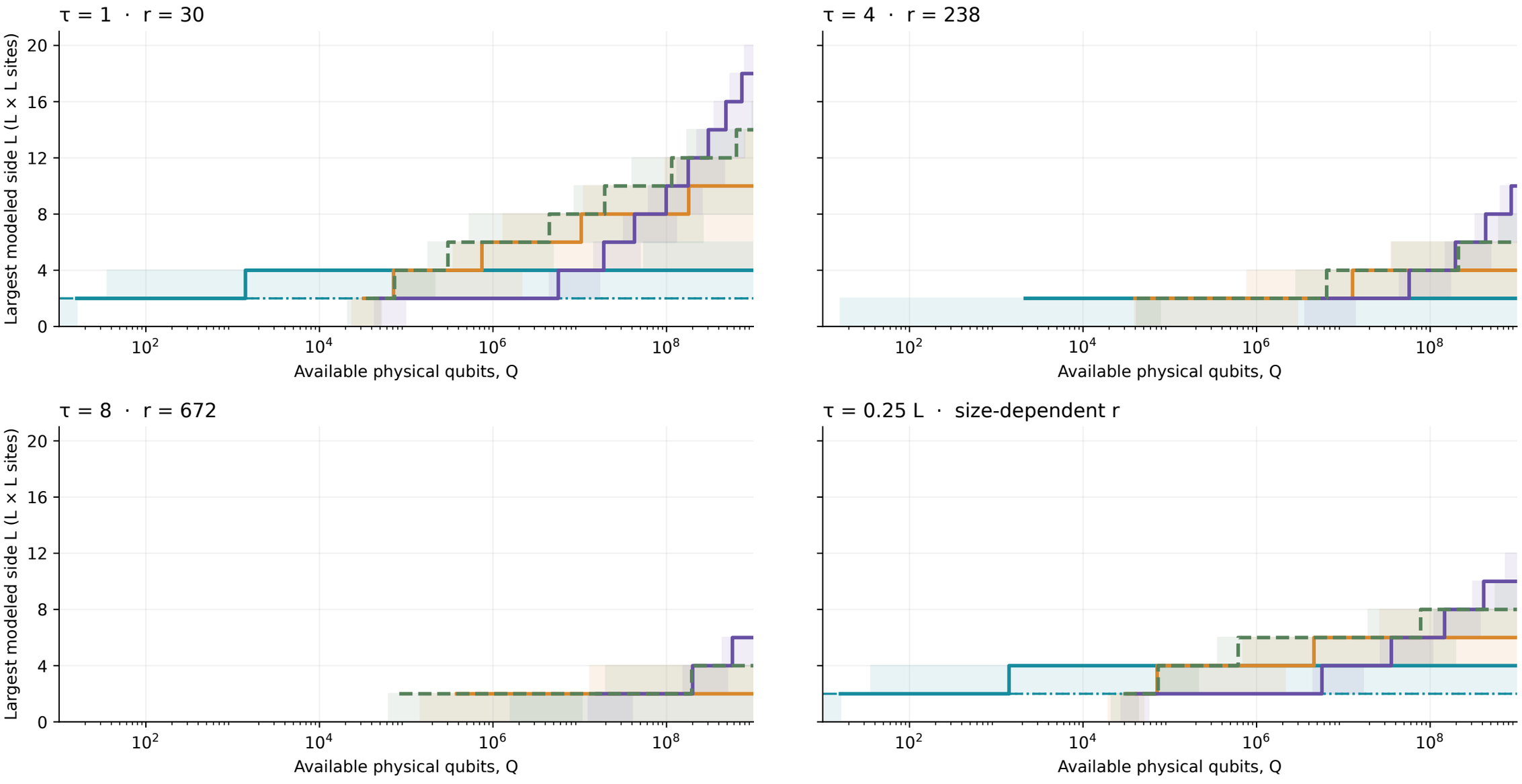

Discussion model: step refinement, mitigation, and hybrid protection
One week · p = 10⁻³ · ε = 0.001 · assumed Trotter error = 0.22 τ³/r², budget = 0.25 ε
τ = 1 · r = 30
τ = 4 · r = 238
τ = 8 · r = 672
τ = 0.25 L · size-dependent r
Largest modeled side L (L × L sites)
Available physical qubits, Q
NISQ · assumed effective mitigation
STAR · analog rotations
FTQC · cultivated T states
Hybrid · analog + fixed T states
NISQ PEC · two-qubit noise
NISQ PEC · compiled-noise proxy
Small-patch-informed scenario, not a large-lattice accuracy certificate. Bands are scenario ranges, including PEC references.
Effective NISQ bias is assumed affordable. PEC uses Hoeffding; other curves use Gaussian planning counts. Full-lattice, even-L estimates.